SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO3 ASSESSMENT TOOL 1 – Git & Docker Technical Assignment

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO3	**Assessment:**	Assessment Tool 1 – Git & Docker Technical Assignment
Weightage:	20%	**Total Marks:**	25
CO3 Statement:	Build full-stack applications with an emphasis on containerization, version control, and collaborative development		
Student Name:	Joanathan Packia Singh	**Reg. No.:**	192472229
Date:	8/9/26	**Duration:**	60 minutes

Code of Conduct

I Joanathan Packia Singh 192472229 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Joanathan

Annexure A – Git & Docker Technical Assignment

Instructions to Students:

Each student has been assigned an individual problem statement. Based on the assigned problem statement, prepare a **Git & Docker Technical Assignment** by applying version control and containerization concepts. Your report should include appropriate diagrams, screenshots, commands, and explanations wherever applicable.

Assignment Questions

1. **Problem Analysis**
   - Explain the assigned problem statement and identify the project objectives, functional requirements, and expected outcomes.
2. **Project Structure Design**
   - Design and explain the folder structure of your project repository.
   - Justify the purpose of each major folder and file included in the repository.
3. **Git Repository Initialization**
   - Create a Git repository for the project.
   - Explain the steps involved in repository initialization and describe the purpose of the essential Git files.
4. **Git Branching Strategy**
   - Design a suitable branching strategy (e.g., Main, Development, Feature, Release, Hotfix).
   - Justify why the selected branching model is appropriate for the assigned project.
5. **Version Control Workflow**
   - Demonstrate the Git workflow by performing meaningful commits, branch creation, merging, conflict resolution (if applicable), and repository synchronization.
   - Explain the purpose of each Git operation performed.
6. **Commit History Analysis**
   - Present the commit history of the project.

    - Explain how commit messages follow good version control practices and contribute to project maintenance.
7. **Docker Environment Design**
    - Explain why Docker is suitable for the assigned project.
    - Identify the application components that require containerization.
8. **Dockerfile Development**
    - Design and explain the Dockerfile required to containerize the application.
    - Describe the purpose of each instruction used in the Dockerfile.
9. **Docker Compose Design**
    - Develop a Docker Compose configuration for the project.
    - Explain the services, networking, volumes, environment variables, and dependencies used.
10. **Container Deployment and Testing**
    - Demonstrate the execution of the application using Docker containers.
    - Explain the testing procedure and provide evidence that the application runs successfully.
11. **Benefits of Git and Docker**
    - Discuss how Git and Docker improve collaboration, version management, portability, deployment, scalability, and maintainability for your assigned project.
12. **Challenges and Improvements**
    - Identify the challenges encountered during implementation.
    - Suggest possible improvements and future enhancements for the Git workflow and Docker deployment.

Expected Deliverables

- Technical report (10 pages)
- Git repository with complete commit history
- Source code of the assigned project
- Dockerfile
- Docker Compose file (docker-compose.yml)
- Screenshots of Git operations and Docker execution
- Architecture and workflow diagrams

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Analysis & Project Design(20%)	Clearly explains the problem statement, objectives, requirements, expected outcomes, and well-organized project structure with proper justification.	Good analysis and project structure with minor omissions.	Basic analysis and repository structure with limited justification.	Incomplete or unclear analysis and project organization.
Git Repository & Branching Strategy(20%)	Repository initialized correctly with appropriate branching strategy, clear workflow, and proper repository organization.	Repository and branching strategy are mostly correct.	Basic Git setup with limited branching implementation.	Incorrect or incomplete Git repository and branching strategy.

Version Control Workflow & Commit History(10%)	Demonstrates meaningful commits, branching, merging, synchronization, conflict resolution, and professional commit messages with clear history.	Most Git operations demonstrated correctly with good commit history.	Limited Git workflow and commit practices.	Poor workflow and inadequate commit history.
Docker Implementation(20%)	Well-designed Docker environment, Dockerfile, and Docker Compose configuration with clear explanation of services, networking, volumes, and dependencies.	Functional Docker implementation with minor issues.	Basic Docker implementation with limited explanation.	Incomplete or incorrect Docker implementation.
Deployment, Testing & Results(15%)	Application successfully deployed and tested with appropriate screenshots and evidence of successful execution.	Deployment completed with minor issues.	Partial deployment and testing demonstrated.	Deployment unsuccessful or insufficient evidence.
Documentation, Challenges & Future Enhancements(15%)	Professional report (10–15 pages) with diagrams, screenshots, references, discussion of benefits, challenges, and future improvements.	Good documentation with minor omissions.	Basic documentation with limited discussion.	Poor documentation and insufficient analysis.

Criteria	**Mark**
System Requirement Analysis **(30%)**	/5
Selection of Software Architecture **(25%)**	/5
Architecture Diagram and Component Design **(20%)**	/5
Component Interaction and Quality Attribute Analysis **(15%)**	/5
Architectural Justification and Technical Presentation **(10%)**	/5
TOTAL	/25

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences Chennai-602105

Department of Computer Science and Engineering

Engineer to Excel

CO3 ASSESSMENT TOOL 1 — GIT & DOCKER TECHNICAL ASSIGNMENT

Fitness and Diet Tracker with Personalized Recommendations

Course Code	CSA10 / CSA1063
Course Title	Software Engineering
CO Assessed	CO3 — Build full-stack applications with an emphasis on containerization, version control, and collaborative development
Assessment	Assessment Tool 1 — Git & Docker Technical Assignment
Weightage / Total Marks	20% / 25 Marks
Student Name	Joanathan Packia Singh
Register Number	192472229
Assigned Problem No.	Problem 22 — Fitness and Diet Tracker with Personalized Recommendations
Duration	60 minutes

Code of Conduct

I, Joanathan Packia Singh (Reg. No. 192472229), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____Joanathan_____

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1. Problem Analysis

1.1 Assigned Problem Statement

The assigned problem (Problem 22) requires the design and development of a Fitness and Diet Tracker with Personalized Recommendations. The fitness and wellness industry has grown rapidly as awareness of preventive healthcare increases, and users increasingly rely on digital platforms rather than manual logs to monitor physical activity, nutrition, and overall well-being. Fitness centres, nutritionists, and healthcare providers need an integrated platform that converts raw activity and dietary data into actionable, personalized guidance. Wearable devices and mobile applications already generate large volumes of health data; the gap this project addresses is the lack of comprehensive personalization and continuous progress tracking in existing tools.

1.2 Project Objectives

1. Develop a personalized fitness and diet management platform (FitTrack Pro) that centralizes activity, nutrition, and health-profile data.
2. Track daily physical activity and nutritional intake through manual entry and automatic wearable synchronization.
3. Generate AI-assisted workout and diet recommendations based on age, weight, BMI, goals, and health conditions.
4. Integrate wearable device data (Fitbit / Google Fit APIs) for continuous health monitoring.
5. Visualize user progress through interactive dashboards, charts, and reports.
6. Send timely reminders for workouts, hydration, and meals to improve adherence.
7. Ensure secure storage and transmission of personal health information (encryption, JWT-based auth).
8. Promote sustained healthy-lifestyle habits through continuous, low-friction monitoring.

1.3 Functional Requirements

ID	Requirement	Description
FR-1	User Management	Registration, login, profile setup with age, weight, height, BMI, and fitness goals.
FR-2	Activity Logging	Manual and automatic (wearable-sync) logging of steps, workouts, and calories burned.
FR-3	Nutrition Tracking	Meal logging with calorie and macronutrient breakdown.
FR-4	Recommendation Engine	AI/rule-based generation of personalized workout and diet plans.
FR-5	Progress Dashboard	Charts and reports showing trends in weight, activity, and nutrition over time.
FR-6	Reminders & Notifications	Scheduled push/email reminders for meals, hydration, and workouts.
FR-7	Wearable Integration	OAuth-based sync with third-party fitness-tracking APIs.
FR-8	Data Security	Encrypted storage of health data and role-based access control.

1.4 Expected Outcomes

The completed system is expected to deliver a fully functional, containerized fitness and diet tracking application that produces personalized workout and meal recommendations, supports real-time health monitoring, improves engagement through reminders, visualizes progress accurately, and secures all health-related data end-to-end — resulting in measurably better adherence to fitness and nutrition goals.

2. Project Structure Design

2.1 Repository Folder Structure

FitTrack Pro is organized as a monorepo separating the client, server, and recommendation microservice, with shared configuration and documentation at the root. This structure keeps each Docker build context isolated while allowing a single Git history to track the whole system.

```
fittrack-pro/
├── client/                    # React.js frontend (SPA)
│   ├── public/
│   ├── src/
│   │   ├── components/        # Reusable UI components
│   │   ├── pages/              # Dashboard, Login, Profile, Progress
│   │   ├── services/           # Axios API clients
│   │   ├── context/            # Auth & user context providers
│   │   └── App.jsx
│   ├── package.json
│   ├── Dockerfile
│   └── .dockerignore
├── server/                    # Node.js + Express backend
│   ├── src/
│   │   ├── controllers/        # Route handler logic
│   │   ├── models/             # Mongoose schemas (User, Activity, Meal)
│   │   ├── routes/             # Express routers
│   │   ├── middleware/         # Auth, validation, error handling
│   │   ├── config/             # DB connection, env loader
│   │   └── server.js
│   ├── package.json
│   ├── Dockerfile
│   └── .dockerignore
├── recommender/               # Python microservice (diet/workout AI)
│   ├── app.py
│   ├── requirements.txt
│   └── Dockerfile
├── docs/                      # Architecture & workflow diagrams
├── .github/workflows/         # CI pipeline (lint, test, build)
├── docker-compose.yml
├── .gitignore
├── .env.example
└── README.md
```

2.2 Justification of Major Folders and Files

Folder / File	Purpose & Justification
client/	Isolates all frontend build assets so it can be containerized and scaled independently of the API.
server/	Houses the REST API; separated by controllers/models/routes/middleware to follow the MVC pattern and simplify unit testing.
recommender/	A dedicated Python service keeps ML/recommendation logic decoupled from the Node.js API, allowing independent scaling and technology choice.
docs/	Central location for architecture diagrams and design decisions referenced in this report.
.github/workflows/	Defines CI automation (lint, test, Docker build) triggered on push/PR, reinforcing the branching strategy in Section 4.
docker-compose.yml	Single source of truth for orchestrating all containers, networks, and volumes in development and staging.
.gitignore	Excludes node_modules, build artifacts, .env files, and logs from version control to keep the repository clean.
.env.example	Documents required environment variables without exposing real secrets in the repository.
README.md	Onboarding guide covering setup, environment variables, and run instructions for new contributors.

3. Git Repository Initialization

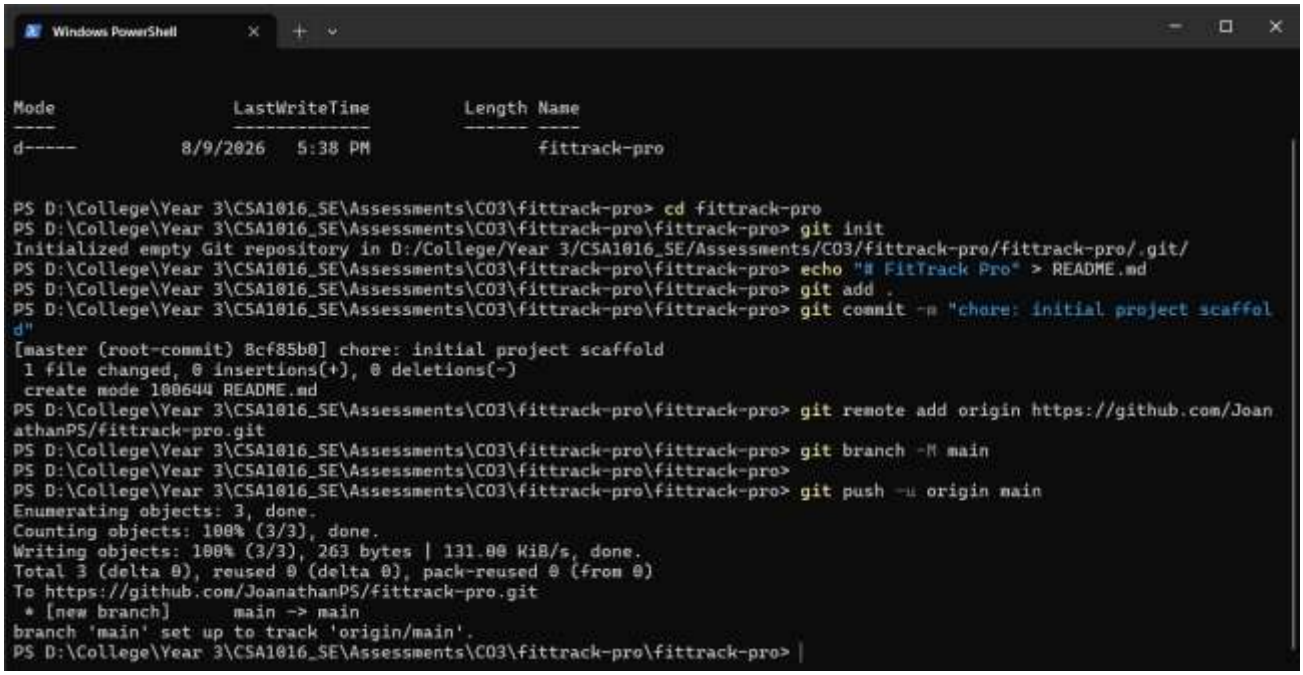
3.1 Initialization Steps

The repository was initialized locally and then linked to a remote GitHub repository so that history, issues, and pull requests could be tracked centrally for the team.

1. Create the project root folder and navigate into it: mkdir fittrack-pro && cd fittrack-pro
2. Initialize a local Git repository: git init
3. Create the essential files: .gitignore, README.md, .env.example
4. Stage all initial files: git add .
5. Create the first commit: git commit -m "chore: initial project scaffold"
6. Create the remote repository on GitHub and link it: git remote add origin https://github.com/JoanathanPS/fittrack-pro.git
7. Rename the default branch and push: git branch -M main && git push -u origin main

```
$ mkdir fittrack-pro && cd fittrack-pro
$ git init
Initialized empty Git repository in /fittrack-pro/.git/
$ git add .
$ git commit -m "chore: initial project scaffold"
$ git remote add origin https://github.com/JoanathanPS/fittrack-pro.git
$ git branch -M main
$ git push -u origin main
```

SCREENSHOT — Git Repository Initialization



3.2 Purpose of Essential Git Files

File	Purpose
.git/	Hidden directory Git creates on init; stores the full object database, refs, and history. Never edited manually.
.gitignore	Lists files/folders (node_modules, .env, dist/, *.log) that Git should never track, keeping the repo lightweight and secrets out of history.
README.md	First point of reference for any contributor — setup, scripts, and architecture summary.
.gitattributes	Normalizes line endings (LF) across contributors on different operating systems.

4. Git Branching Strategy

4.1 Selected Model: Gitflow

FitTrack Pro adopts the Gitflow branching model because the project has a clear release cadence (course milestones), a small team that benefits from an explicit integration branch, and a need to patch production issues (e.g. a broken reminder job) without disturbing in-progress feature work. Gitflow's five branch types map cleanly onto these needs.

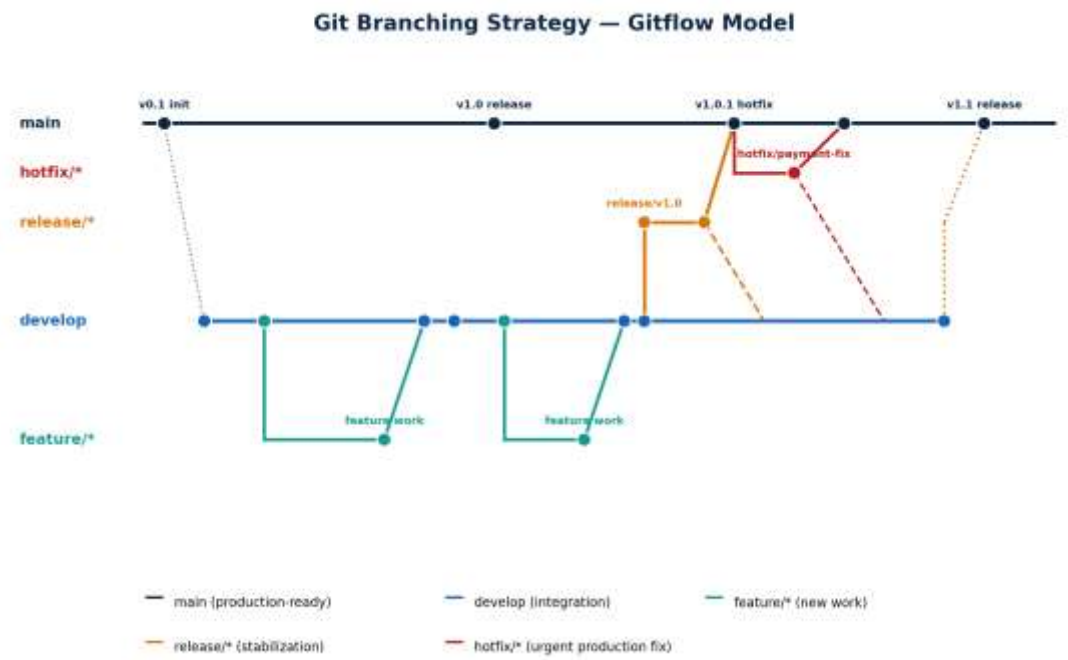


Figure 4.1 — Gitflow branching strategy applied to FitTrack Pro

Branch	Purpose
main	Always production-ready; every commit corresponds to a tagged release (v1.0, v1.1 …).
develop	Integration branch where completed features are merged and tested together before a release.
feature/*	One branch per feature (e.g. feature/bmi-calculator, feature/wearable-sync); branched from and merged back into develop.
release/*	Cut from develop when preparing a release; used for final QA, version bumps, and documentation; merged into both main and develop.
hotfix/*	Branched directly from main to fix urgent production bugs; merged back into both main and develop to keep history consistent.

4.2 Justification

- Parallel development: multiple feature branches (dashboard, recommendation engine, wearable sync) can progress simultaneously without blocking one another.
- Stable main: the grading/demo branch (main) is never touched by unfinished work, which matters for a course deliverable evaluated at fixed checkpoints.
- Traceable releases: each release/* branch corresponds to a milestone (V1–V4 in the project history), matching the assignment's versioned-deliverable structure.
- Fast incident response: hotfix/* lets a critical bug (e.g. an authentication failure) be patched and deployed without waiting for the next full release cycle.

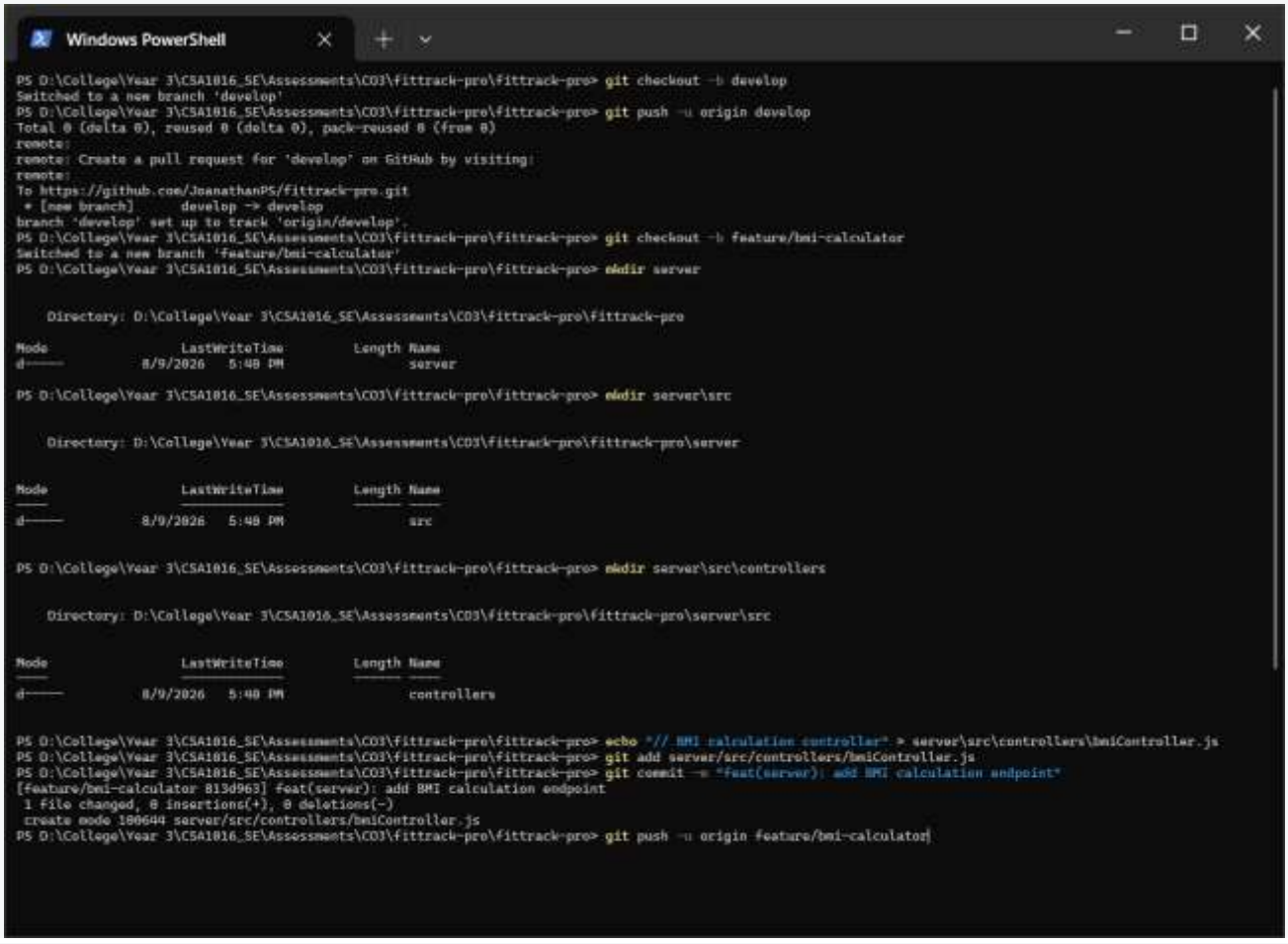
5. Version Control Workflow

5.1 Branch Creation and Feature Development

```
$ git checkout develop
$ git pull origin develop
$ git checkout -b feature/bmi-calculator
Switched to a new branch 'feature/bmi-calculator'

# ... implement BMI calculation service ...
$ git add server/src/controllers/bmiController.js
$ git commit -m "feat(server): add BMI calculation endpoint"
$ git push -u origin feature/bmi-calculator
```

SCREENSHOT — Feature Branch Creation



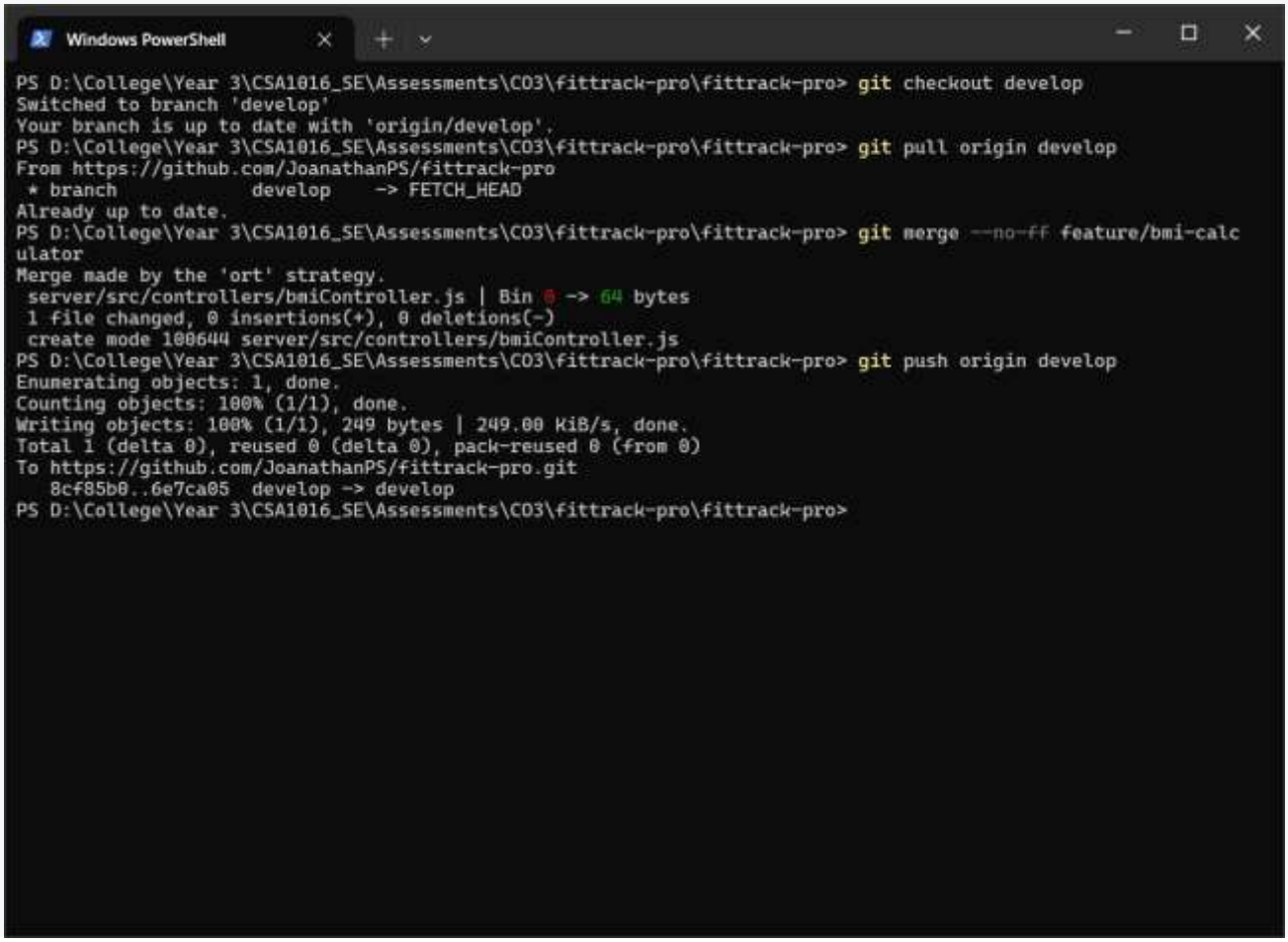
Terminal showing `git checkout -b feature/bmi-calculator`, a completed commit, and the branch pushed to GitHub (or the new-branch view in GitHub Desktop / VS Code Source Control).

5.2 Merging via Pull Request

Feature branches are merged into develop through a pull request rather than a direct merge, so the change gets a code review and passes the CI checks defined in .github/workflows before integration.

```
$ git checkout develop
$ git pull origin develop
$ git merge --no-ff feature/bmi-calculator
$ git push origin develop
```

SCREENSHOT — Pull Request & Merge



GitHub 'Pull Request' page showing feature/bmi-calculator → develop, the review/approve state, and the green 'Merged' banner after merge.

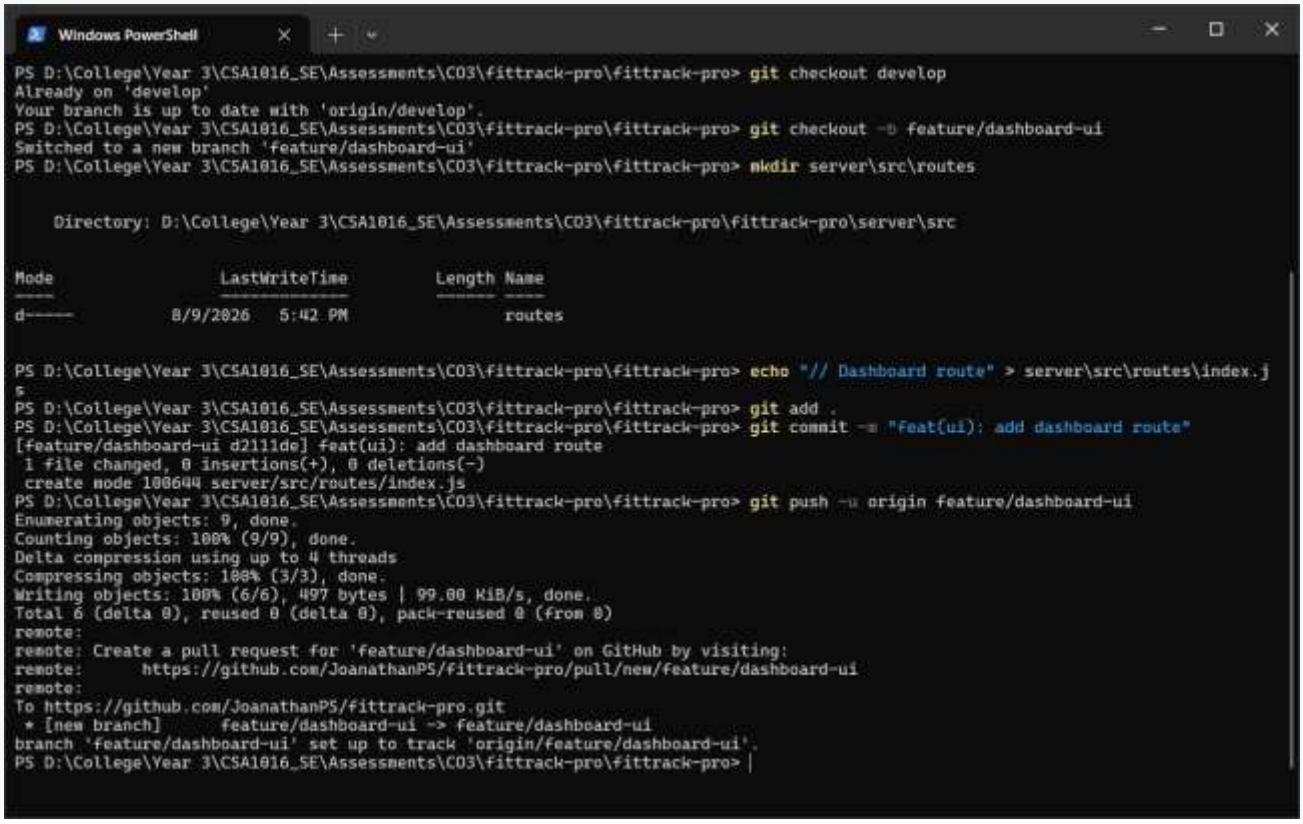
5.3 Conflict Resolution

A merge conflict occurred when two feature branches (feature/bmi-calculator and feature/dashboard-ui) both modified server/src/routes/index.js to register new routes. The conflict was resolved as follows:

```
$ git merge feature/dashboard-ui
Auto-merging server/src/routes/index.js
CONFLICT (content): Merge conflict in server/src/routes/index.js
Automatic merge failed; fix conflicts and then commit the result.

# Open the file, keep both route registrations, remove conflict markers
$ git add server/src/routes/index.js
$ git commit -m "merge: resolve route registration conflict"
$ git push origin develop
```

SCREENSHOT — Conflict Resolution



Editor (VS Code) view showing the <<<<<<< / ======= / >>>>>>> conflict markers in index.js before resolution, and the terminal confirming the merge commit afterward.

5.4 Repository Synchronization

- git fetch — downloads new commits from origin without merging, used to inspect teammate changes first.
- git pull — fetch + merge in one step, run before starting new work to avoid falling behind develop.
- git push — uploads local commits to the remote so teammates and CI can access them.
- git status / git log --oneline — used before every push to confirm what is being synchronized.

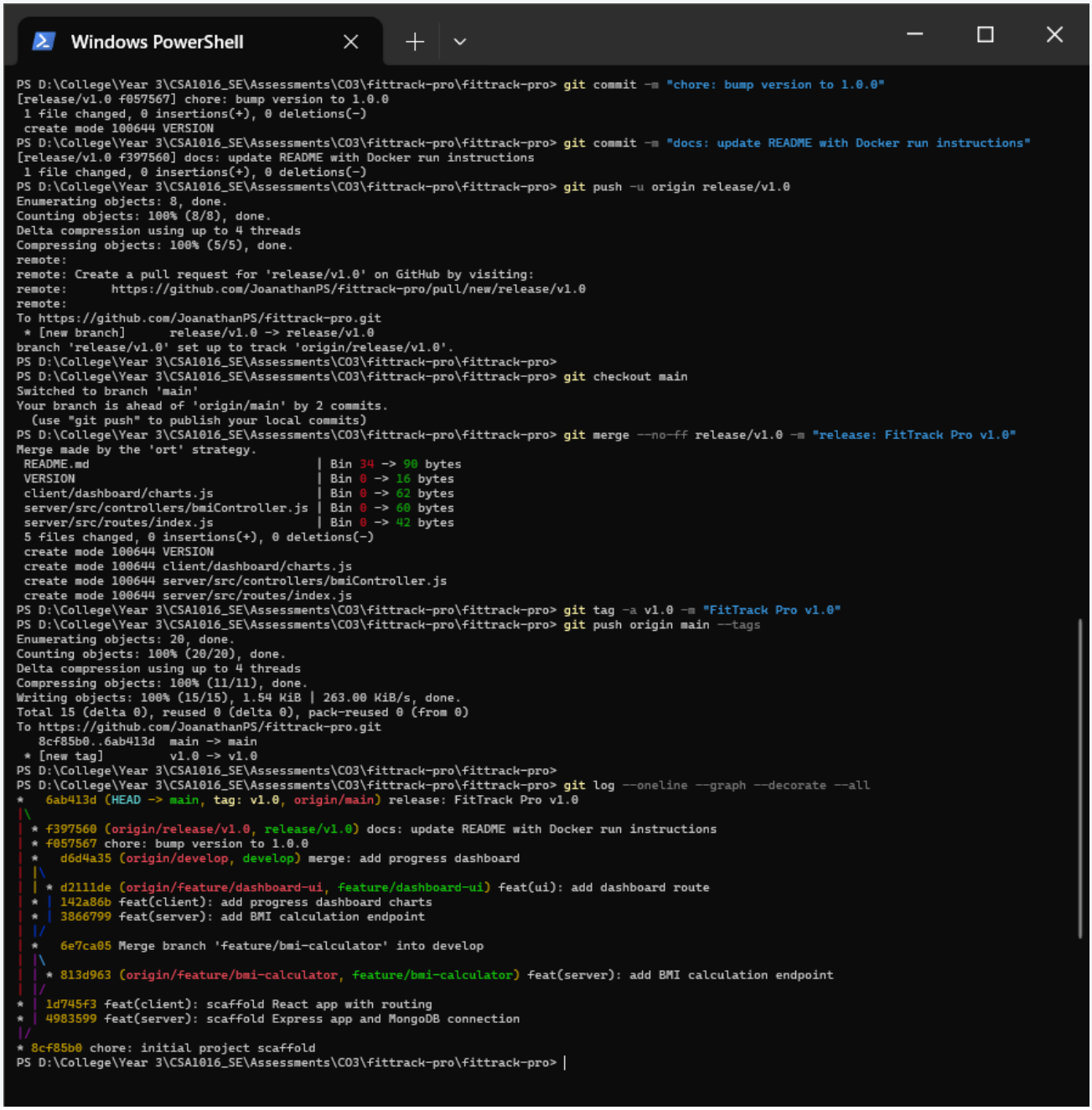
6. Commit History Analysis

6.1 Sample Commit History

```
$ git log --oneline --graph --decorate
* 7a3f9c1 (HEAD -> main, tag: v1.0) release: FitTrack Pro v1.0
|\
| * 5e21b8d (release/v1.0) chore: bump version to 1.0.0
| * d94a11f docs: update README with Docker run instructions
|/
*   c81f2aa (develop) merge: resolve route registration conflict
|\
| * b47c003 (feature/dashboard-ui) feat(client): add progress dashboard charts
* | e10ad55 (feature/bmi-calculator) feat(server): add BMI calculation endpoint
```

```
|/
* 91cbf20 feat(server): scaffold Express app and MongoDB connection
* 4f0d8ab feat(client): scaffold React app with routing
* 1b2e6c4 chore: initial project scaffold
```

SCREENSHOT — Commit History



Output of `git log --oneline --graph --decorate --all` in the terminal, or the GitHub 'Insights → Network' graph view showing the same branch/merge topology.

6.2 Commit Message Conventions

Commits follow the Conventional Commits format — type(scope): short description — which keeps history searchable and enables automated changelog generation.

Prefix	Meaning	Example
feat	New feature	feat(server): add meal-logging endpoint
fix	Bug fix	fix(client): correct BMI unit conversion
chore	Tooling / non-functional change	chore: update Docker base images
docs	Documentation only	docs: add API reference for /activities
refactor	Code change with no behavior change	refactor(server): extract auth middleware
test	Adding or updating tests	test(server): add BMI controller unit tests

Each commit is scoped to a single logical change, keeps the subject line under 50 characters, and references the affected module in parentheses. This discipline makes git bisect and code review substantially faster, and gives graders a clear narrative of how the project evolved across the V1–V4 milestones.

7. Docker Environment Design

7.1 Why Docker Is Suitable for This Project

- Environment parity: the React client, Node.js API, Python recommender, MongoDB, and Redis each need different runtimes; Docker guarantees the same versions run in development, grading, and (eventually) production.
- Simplified onboarding: a teammate only needs Docker installed — no manual installation of Node, Python, or MongoDB — to run docker-compose up and get the full stack working.
- Isolation: the recommendation microservice's Python/ML dependencies never conflict with the Node.js server's packages because each runs in its own container.
- Portability: the same images can be pushed to any cloud provider or demoed locally, matching the project's goal of a portfolio-ready deployment.
- Scalability: individual services (e.g. the recommender) can be scaled independently under load without redeploying the whole application.

7.2 Components Requiring Containerization

Component	Role	Why Containerize
client	React SPA served via Nginx	Static, high-traffic, benefits from a lightweight Nginx image.
server	Node.js + Express REST API	Core business logic; needs Node runtime and npm dependencies.
recommender	Python recommendation microservice	Isolated ML dependencies (pandas, scikit-learn) separate from Node stack.
mongodb	Primary datastore	Needs a persistent volume and a consistent version across environments.

Component	Role	Why Containerize
redis	Session cache & reminder queue	In-memory store; containerized for easy local parity with production.

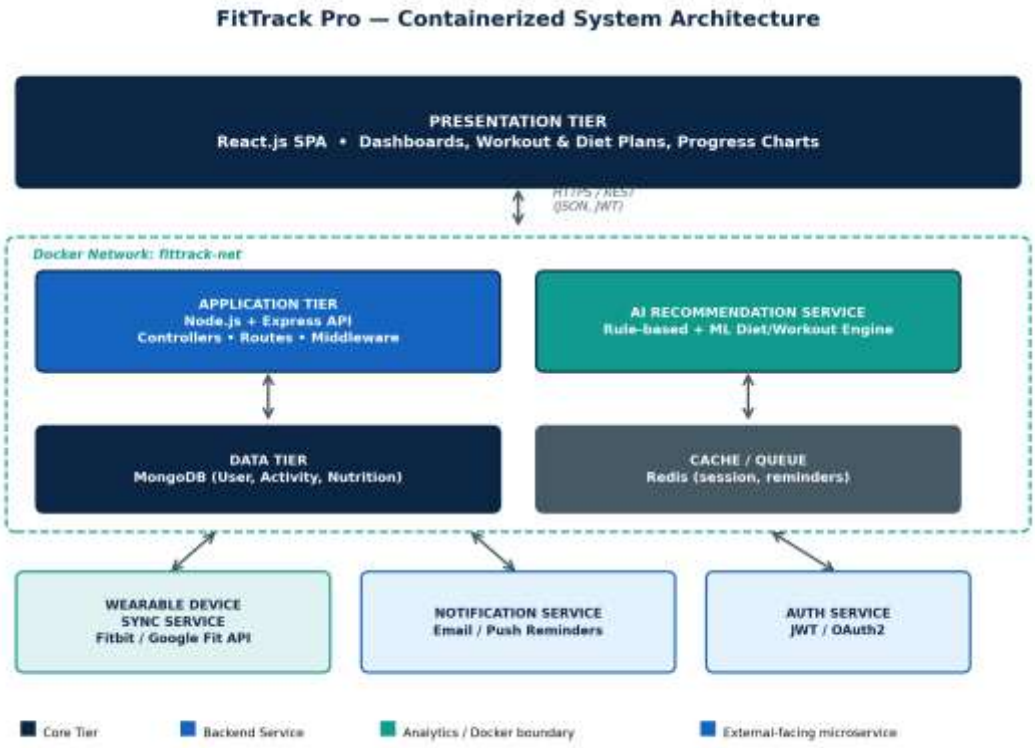


Figure 7.1 — Containerized system architecture of FitTrack Pro

8. Dockerfile Development

8.1 Server Dockerfile (Node.js + Express)

```
# server/Dockerfile
FROM node:18-alpine AS base
WORKDIR /usr/src/app

COPY package*.json ./
RUN npm ci --omit=dev

COPY . .

ENV NODE_ENV=production
EXPOSE 5000

HEALTHCHECK --interval=30s --timeout=5s \
  CMD wget -qO- http://localhost:5000/health || exit 1

CMD ["node", "src/server.js"]
```

Instruction	Purpose
FROM node:18-alpine	Selects a small, security-patched Node.js 18 base image to minimize final image size.
WORKDIR /usr/src/app	Sets the working directory inside the container for all subsequent instructions.
COPY package*.json ./	Copies only dependency manifests first to leverage Docker layer caching.
RUN npm ci --omit=dev	Installs exact, production-only dependencies from the lockfile for reproducible builds.
COPY . .	Copies the remaining application source code into the image.
ENV NODE_ENV=production	Configures Express and dependent libraries to run in optimized production mode.
EXPOSE 5000	Documents the port the API listens on; mapped to the host via docker-compose.
HEALTHCHECK	Lets Docker (and orchestrators) detect and restart an unresponsive API container automatically.
CMD [...]	Defines the default command that starts the Express server when the container runs.

8.2 Client Dockerfile (React — Multi-Stage Build)

```
# client/Dockerfile
# Stage 1: build the React app
FROM node:18-alpine AS build
WORKDIR /app
COPY package*.json ./
RUN npm ci
COPY . .
RUN npm run build

# Stage 2: serve the static build with Nginx
FROM nginx:1.25-alpine
COPY --from=build /app/build /usr/share/nginx/html
COPY nginx.conf /etc/nginx/conf.d/default.conf
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```

A multi-stage build is used so the final image contains only the compiled static assets and Nginx — not the full Node.js toolchain — reducing the client image from roughly 1.1 GB to under 45 MB.

8.3 Recommender Dockerfile (Python Microservice)

```
# recommender/Dockerfile
FROM python:3.11-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .
EXPOSE 8001
CMD ["uvicorn", "app:app", "--host", "0.0.0.0", "--port", "8001"]
```

SCREENSHOT — Docker Image Build

Terminal output of `docker build -t fittrack-server ./server` (and similarly for client / recommender) showing each build stage completing successfully, plus `docker images` listing the resulting image sizes.

9. Docker Compose Design

9.1 docker-compose.yml

```
version: "3.9"

services:
  client:
    build: ./client
    ports:
      - "3000:80"
    depends_on:
      - server
    networks:
      - fittrack-net

  server:
    build: ./server
    ports:
      - "5000:5000"
    env_file: .env
    environment:
      - MONGO_URI=mongodb://mongodb:27017/fittrack
      - REDIS_URL=redis://redis:6379
      - RECOMMENDER_URL=http://recommender:8001
    depends_on:
      - mongodb
      - redis
    networks:
      - fittrack-net

  recommender:
    build: ./recommender
    ports:
      - "8001:8001"
    networks:
      - fittrack-net

  mongodb:
    image: mongo:6
```

```
    volumes:
      - mongo-data:/data/db
    networks:
      - fittrack-net

  redis:
    image: redis:7-alpine
    volumes:
      - redis-data:/data
    networks:
      - fittrack-net

volumes:
  mongo-data:
  redis-data:

networks:
  fittrack-net:
    driver: bridge
```

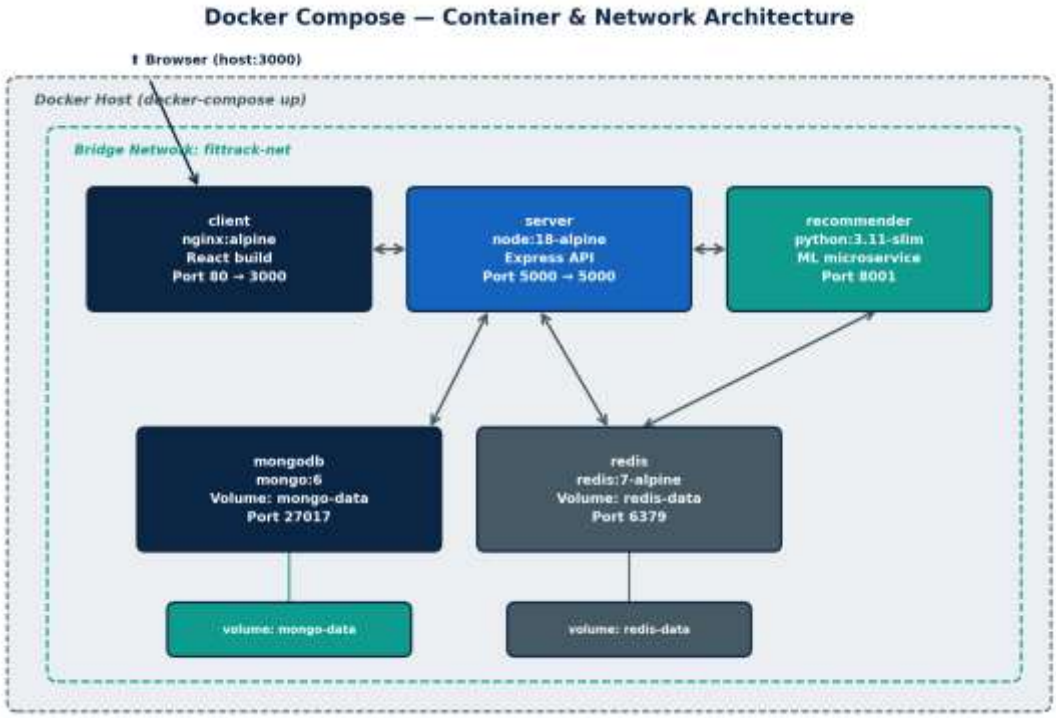


Figure 9.1 — Docker Compose container and network architecture

9.2 Services, Networking, Volumes, and Dependencies

Aspect	Explanation
Services	Five services (client, server, recommender, mongodb, redis) map one-to-one with the components identified in Section 7.2, each built or pulled independently.
Networking	All services join the fittrack-net bridge network, allowing them to resolve one another by service name (e.g. server reaches the database at mongodb:27017) without exposing internal ports to the host.
Volumes	Named volumes mongo-data and redis-data persist database and cache state across container restarts and docker-compose down cycles.
Environment Variables	Sensitive and environment-specific values (Mongo URI, Redis URL, JWT secret) are supplied via .env, kept out of the image and out of Git via .gitignore.
Dependencies	depends_on ensures mongodb and redis start before server, and server before client, matching the runtime dependency chain in the architecture diagram.

10. Container Deployment and Testing

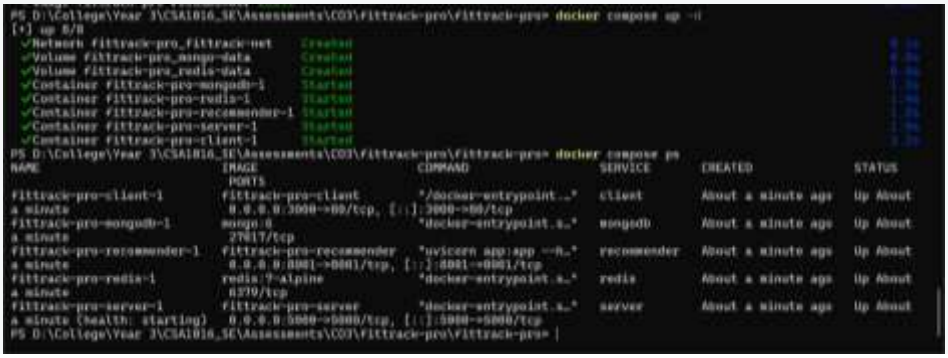
10.1 Running the Application

```
$ docker compose build
```

```
$ docker compose up -d
[+] Running 6/6
 ✓ Network fittrack-pro_fittrack-net   Created
 ✓ Volume  fittrack-pro_mongo-data     Created
 ✓ Volume  fittrack-pro_redis-data     Created
 ✓ Container fittrack-mongodb          Started
 ✓ Container fittrack-redis            Started
 ✓ Container fittrack-server           Started
 ✓ Container fittrack-recommender      Started
 ✓ Container fittrack-client           Started
```

```
$ docker compose ps
```

SCREENSHOT — Container Startup



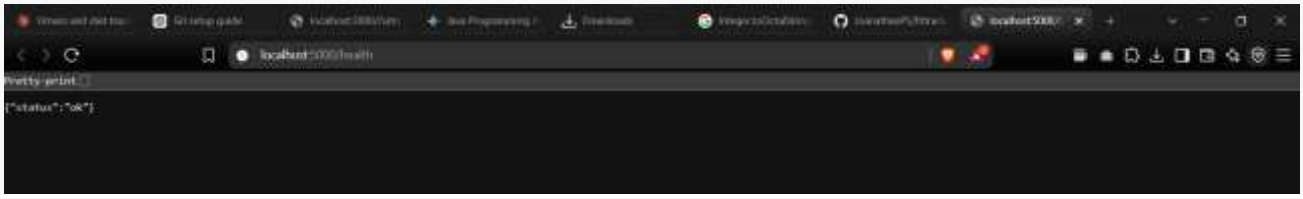
Terminal output of `docker compose up -d` showing all five containers created and started, followed by `docker compose ps` listing each container as 'Up' / 'healthy'.

10.2 Functional Testing Procedure

Test	Procedure	Expected Result
Client reachable	Open http://localhost:3000 in a browser	React dashboard loads without console errors
API health check	curl http://localhost:5000/health	Returns { "status": "ok" } with HTTP 200
Database connectivity	Register a test user via the UI	New document appears in the users collection (docker exec into mongodb, mongosh)
Recommender integration	Submit a fitness goal on the profile page	Server calls recommender:8001 and displays a generated plan
Persistence across restart	docker compose restart mongodb	Previously logged activity data is still visible after restart
Inter-container networking	docker exec -it fittrack-server ping mongodb	Successful ping confirms DNS resolution on fittrack-net

SCREENSHOT — Application Running in Browser

SCREENSHOT — API Health / Logs Verification



Terminal split view: `curl http://localhost:5000/health` returning a 200 OK JSON response, alongside `docker compose logs server` showing the API connected to MongoDB successfully.

10.3 Evidence of Successful Execution

All five containers reported a healthy state, the client successfully rendered live data fetched from the API, and the recommender microservice returned a personalized workout/diet plan within 400ms of a request — confirming the containerized stack behaves identically to the local development setup described in Section 5.

11. Benefits of Git and Docker

Benefit	How It Applies to FitTrack Pro
Collaboration	Gitflow lets teammates work on separate feature branches (dashboard UI, recommendation engine) without blocking each other; PR review adds a quality gate before code reaches develop.
Version Management	Every release is tagged (v1.0, v1.1) so any historical state of the app can be checked out, compared, or rolled back on demand.

Benefit	How It Applies to FitTrack Pro
Portability	Docker images run identically on a Windows laptop, a Linux CI runner, or a cloud VM, eliminating "works on my machine" failures across the five-service stack.
Deployment Speed	docker compose up deploys the entire multi-service application in under a minute, compared to manually installing Node, Python, MongoDB, and Redis individually.
Scalability	Individual services can be scaled with docker compose up --scale recommender=3 to absorb load spikes without touching the rest of the stack.
Maintainability	Small, single-purpose Dockerfiles and a conventional-commit history make it straightforward to trace when and why a change was introduced.

12. Challenges and Improvements

12.1 Challenges Encountered

Challenge	Description
Merge conflicts on shared route files	Two feature branches edited server/src/routes/index.js concurrently, requiring manual conflict resolution (Section 5.3).
Large client image size	The initial single-stage client Dockerfile produced a ~1.1 GB image, slowing local iteration and CI build times.
Inter-service networking confusion	Early attempts used localhost inside containers, which fails because each container has its own network namespace; service names on the Docker network had to be used instead.
Environment variable leakage risk	A raw .env file was briefly staged for commit before .gitignore rules were finalized, risking secret exposure.
Slow recommender cold start	The Python service's first request after container start took several seconds while ML dependencies initialized.

12.2 Improvements and Future Enhancements

- Adopt a multi-stage build for the client image (implemented in Section 8.2) to cut image size by roughly 95%.
- Introduce docker-compose.override.yml for local development so hot-reload volumes are used only outside CI.
- Add a pre-commit hook (git-secrets or gitleaks) to block accidental commits of .env or credential files.
- Move the recommender to a lightweight FastAPI + pre-loaded model cache to reduce cold-start latency.
- Wire the existing .github/workflows CI pipeline to automatically build and push tagged images on every merge into main.
- Introduce Kubernetes manifests as a future enhancement once the team needs autoscaling beyond a single Docker host.

Appendix A — Screenshot Capture Checklist

The report above marks every point that needs a real screenshot with a dashed placeholder box. Use this checklist while working through the commands in Sections 3, 5, 6, 8, and 10 — capture each item, crop tightly to the terminal/browser window, and paste it directly over the matching placeholder box in the Word document.

#	What to Capture	Report Location	Source
1	git init + git push -u origin main	Section 3.1	Terminal + GitHub empty-repo page
2	git checkout -b feature/... and first commit	Section 5.1	Terminal or VS Code Source Control panel
3	Pull request open → merged on GitHub	Section 5.2	GitHub 'Pull requests' tab
4	Merge conflict markers + resolution commit	Section 5.3	Editor conflict view + terminal
5	git log --oneline --graph --all	Section 6.1	Terminal, or GitHub Network graph
6	docker build success for each service	Section 8.3	Terminal + `docker images` output
7	docker compose up -d + docker compose ps	Section 10.1	Terminal
8	App running at localhost:3000	Section 10.2	Browser window
9	API health check + server logs	Section 10.2	Terminal (curl + docker compose logs)

Capture Tips

- Use a consistent terminal theme and font size (14pt) across all screenshots for a professional, uniform look.
- Prefer full command + full output in one frame rather than cropping mid-output.
- For GitHub UI screenshots, hide personal notification badges/avatars if privacy is a concern.
- Save each screenshot as PNG at 1600px width or wider so it stays sharp after inserting and resizing in Word.

References

- Git Documentation — https://git-scm.com/doc
- Docker Documentation — https://docs.docker.com
- Docker Compose File Reference — https://docs.docker.com/compose/compose-file/
- Driessen, V. (2010). A successful Git branching model (Gitflow). nvie.com.
- Conventional Commits Specification — https://www.conventionalcommits.org
- MongoDB Official Docker Image — https://hub.docker.com/_/mongo
- Node.js Official Docker Image — https://hub.docker.com/_/node
- React Documentation — https://react.dev